

Chapter 03 Raw Textbook Content

Textbook of Minimally Invasive Spine Surgery (Springer Nature 2026) | Full Raw Text Extraction

Textbook of Minimally Invasive Spine Surgery, Concepts and Surgical Techniques

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3. Clinical Evidence of Minimally Invasive Spine Surgery: A Review

Department of Orthopedics, Ruijin Hospital, Shanghai Jiaotong University School of Medicine, Shanghai, People's Republic of China

In recent decades, the emergence of minimally invasive spinal surgery (MISS) has transformed the management of spinal disorders through advanced techniques that enable precise intervention while preserving surrounding tissues [1]. Unlike traditional open spine surgery—which requires extensive tissue dissection via larger incisions—MISS utilizes miniature incisions, specialized access working portals and extended instruments, and advanced imaging technologies to address spinal pathology [2]. These approaches are indicated for neural decompression, spinal oncology, traumatic injuries, and fusion procedures, achieving analogous therapeutic outcomes to open surgery with demonstrably reduced iatrogenic

3.1 Lumbar

3.1.1 Lumbar Disc Herniation and Discectomy

Lumbar disc herniation (LDH)—a prevalent clinical condition that causes symptomatic radiculopathy—is a leading cause of disability. Patients who failed with conservative management often require a discectomy [6]. Conventional open discectomy involves a midline lumbar incision, paraspinal muscle dissection, laminotomy, and ligamentum flavum resection to expose the dura and nerve roots for herniated nucleus pulposus removal. In contrast, minimally invasive surgery (MIS) techniques preserve paraspinal musculature, laminae, facet joints, and ligamentous structures, thereby reducing segmental destabilization and accelerating recovery. These approaches offer distinct advantages for highly migrated, far-lateral, and upper lumbar disc herniations. Current MIS discectomy options include tubular retractor–assisted microdiscectomy, uniportal endoscopic (uEndo) discectomy, and biportal endoscopic discectomy.

3.1.1.1 Tubular Retractor–Assisted Microdiscectomy

The inaugural microdiscectomy was performed by Kravenbühl and Ya argil in 1968 [7]. Subsequent innovations included automated percutaneous lumbar discectomy and laser-assisted discectomy [8, 9]. A period of significant progress in tubular access systems followed, leading to the advent of minimally invasive surgery using